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Vietnamese Automatic Speech Recognition Utilizing Audio and Visual Data

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INTRODUCTION

Problem: While humans use both audio and visual cues to understand speech, most audio-visual speech recognition datasets are for English. This creates a significant gap for developing effective Automatic Speech Recognition (ASR) systems for Vietnamese.

Solution:

- **VASR Dataset:** The first large-scale Vietnamese Audio-Visual Speech Recognition dataset, containing 1,045 hours of multi-view video.
- **ViAVSP-LLM Framework:** A novel model that integrates an AV-HuBERT encoder with a VinaLLaMA decoder to process both audio and visual speech data.

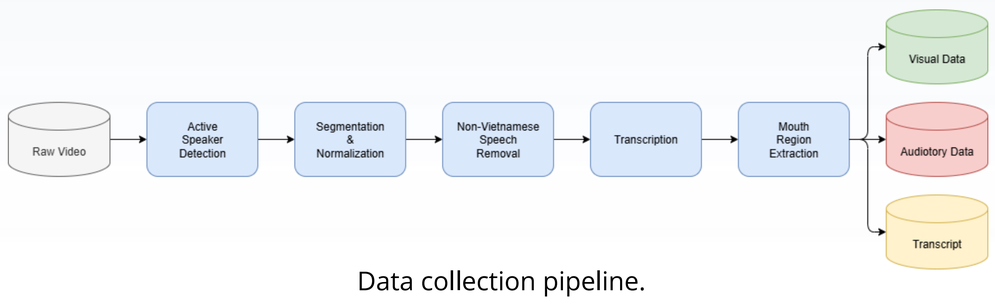
Key Contributions:

- Creation of the extensive VASR dataset.
- Development and optimization of the ViAVSP-LLM framework.
- Demonstration that adding visual data and a large language model significantly improves Vietnamese speech recognition accuracy.

METHOD

VASR Dataset

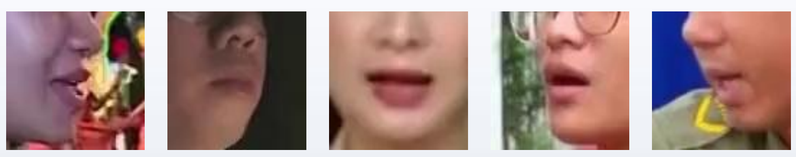
Data Sourcing: Videos were collected from online platforms like YouTube and TikTok.



Data collection pipeline.

Processing Pipeline:

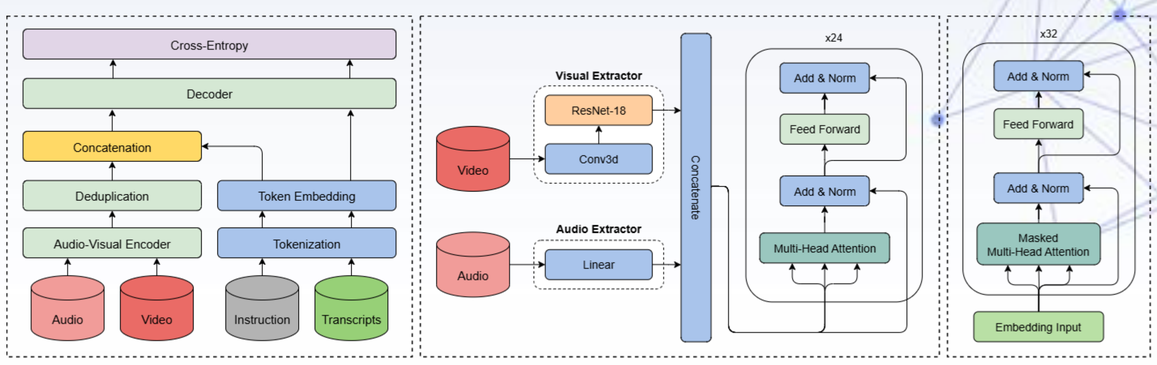
1. **Active Speaker Detection:** Used TalkNet to find segments with active speakers.
2. **Segmentation & Normalization:** Clips were cut to 3 seconds, with audio normalized to 16,000 Hz and video to 25 FPS.
3. **Filtering & Transcription:** Non-Vietnamese speech was removed using VoxLingua107, and the remaining audio was transcribed.
4. **Visual Focus:** The mouth region was cropped to concentrate on visual speech cues.



Examples of different camera views.

Dataset Characteristics: The final dataset includes 1,045 hours of video, covers 11 topics, and features varied camera angles and lighting to simulate real-world conditions.

ViAVSP-LLM



Left: Overview of ViAVSP-LLM architecture. **Middle and Right:** Details of Audio-Visual Encoder and Decoder, respectively.

1. Audio-Visual Encoder (AV-HuBERT):

- This component processes the audio and visual streams.
- **Audio Extractor:** A linear projection layer processes the audio signal.
- **Visual Extractor:** A 3D Convolutional layer and a ResNet-18 model extract features from the mouth-region video, capturing temporal relationships.
- The features from both modalities are then processed by a Transformer-based encoder.

2. Deduplication: A K-means clustering technique is used to group similar audio-visual features, reducing the data sequence length to make it manageable for the decoder without losing essential information.

3. Decoder (VinaLLaMA):

- The reduced features are fed into a VinaLLaMA-based decoder.
- This part of the model uses Masked Multi-Head Attention to generate the text transcription auto-regressively.
- To manage computational costs, Low-Rank Adaptation (LoRA) was used during fine-tuning.

RESULT

Main Quantitative Finding: The final ViAVSP-LLM model achieved a **Word Error Rate (WER) of 12.03%** on the test set. This performance is a significant improvement over audio-only models.

Qualitative Analysis:

- **Successes:** The model showed high accuracy in aligning transcriptions with the actual speech, thanks to the contextual understanding from the LLM.
- **Challenges:**
  - Performance decreased with poor audio quality (e.g., background noise, rapid speech).
  - The model struggled with untranslated English words (like "comment" or "livestream") mixed into Vietnamese speech, as the VinaLLaMA decoder was trained only on Vietnamese.